

4.19 Quiz: Probability (Algebra)

1. Events A and B are independent. It is given that $P(A) = 2P(B)$ and $P(A \cup B) = 0.72$.

Let $x = P(B)$.

(a) Show that $2x^2 - 3x + 0.72 = 0$. [3]

(b) Hence find $P(B)$. [2]

(c) Find $P(A \cap B)$. [1]

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2. In a school, 80 students were surveyed about activities A and B .

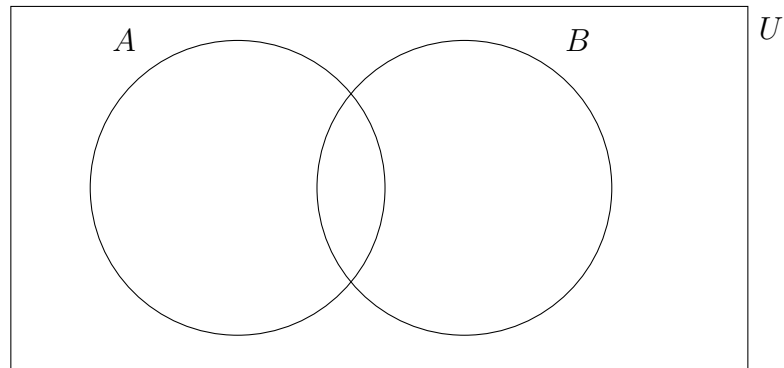
- $P(A) = 0.5$
- $P(B \mid A) = 0.3$
- $P(A' \cap B') = 0.2$

(a) Find $P(A \cap B)$. [2]

(b) Find $P(B)$. [2]

(c) Complete the Venn diagram below, showing the number of students in each region. [2]

(d) Determine whether events A and B are independent. Justify your answer. [2]



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